



YouTube Analytics Coursework

Problem Statement & Objective

Business Problem: Predict which YouTube videos are likely to go viral and estimate their view counts.

Objectives:

1. Explore video engagement metrics and understand patterns.
2. Use Linear Regression to predict views (regression problem).
3. Use Logistic Regression to classify viral vs non-viral videos (classification problem).
4. Apply hypothesis testing to validate business assumptions.
5. Preprocess data using scaling and encoding.
6. Evaluate models using MSE, RMSE, MAE, R^2 , Confusion Matrix, Precision, Recall, F1-score, ROC Curve, ROC-AUC.
7. Derive actionable business insights for content strategy.

Task 1: Load Dataset & Define Targets

Instructions:

- Load [USvideos.csv](#) from Kaggle or any public source.
- Inspect the data using [head\(\)](#), [info\(\)](#), [describe\(\)](#).
- Identify **regression target** → [views](#)
- Identify **classification target** → viral if views > top 25% quantile.
- Create a new column [viral](#) for classification.

Expected outputs:

- Display of dataset shape, first 5 rows.
- Correctly created [viral](#) column (binary).

Task 2: Explore Distributions and Relationships

Instructions:

- Plot histogram of views.
- Plot scatterplot: likes vs views.
- Compute correlation matrix for numeric features:
[views](#), [likes](#), [dislikes](#), [comment_count](#), [duration](#), [title_length](#), [days_since_publish](#).
- Students should [comment on relationships](#) observed.



Expected outputs:

- Correct plots and correlation matrix.
- Short textual observation: e.g., "Likes show strong positive correlation with views."

Task 3: Hypothesis Testing

Instructions:

- Test the hypothesis: "Videos with higher likes, dislikes, or comment counts have significantly higher views."
- Use t-test to calculate p-values for each feature.
- Interpret the p-values and state whether the hypothesis is supported.

Expected Output:

- p-values for likes, dislikes, and comment counts.
- Clear textual interpretation: e.g., $p < 0.05 \rightarrow$ reject null hypothesis; otherwise, fail to reject.

Task 4: Preprocess Features

Instructions (Student Version):

1. Select relevant numeric features: `likes`, `dislikes`, `comment_count`, `title_length`, `days_since_publish`
2. Encode categorical variables: `channel_title`, `category_id` using **one-hot encoding**.
3. Scale numeric features using **StandardScaler**.
4. Split the dataset into training and testing sets (80%-20%).

Expected Outputs:

- Properly scaled numeric features.
- Encoded categorical features combined with numeric features.
- Correct train-test split for both regression and classification targets.

Task 5: Linear Regression

Instructions:

- Train a **Linear Regression** model using training data.
- Predict **views** on the test set.
- Evaluate using **MSE, RMSE, MAE, R^2** .
- Visualize **feature importance / coefficients**.

Expected outputs:

- Regression metrics printed.
- Feature importance plot.
- Brief comment on which features influence views most.

Task 6: Logistic Regression

Instructions:

- Train **Logistic Regression** on classification target **viral**.
- Predict on test set.
- Evaluate using **Accuracy, Precision, Recall, F1-score, ROC-AUC**.
- Plot **confusion matrix and ROC curve**.

Expected outputs:

- Confusion matrix plot.
- ROC curve plot.
- Printed evaluation metrics.
- Short interpretation: which features influence virality most.

Task 7: Cross-Validation and Scaling Comparison

Instructions:

- Perform **cross-validation** (5 folds) on Linear Regression (**views**) and Logistic Regression (**viral**).
- Compare scaling techniques: StandardScaler vs MinMaxScaler.
- Visualize and interpret **feature importance** for logistic regression.

**Expected outputs:**

- Cross-validation scores (mean and variance).
- Comparison of scaled datasets.
- Feature importance plot and textual interpretation.

Task 8: Report & Insights

Instructions:

- Summarize results in a short report:
 - Regression model performance and key predictors of views.
 - Classification model performance and key predictors of viral videos.
 - Insights from hypothesis testing.
 - Business recommendations (e.g., optimal video length, engagement strategies).

Expected outputs:

- Well-structured report with tables, plots, and observations (.py format).
- Clear actionable insights based on model results.